men and traveling (so the report says) at 4 leagues an hour on the canal.

The boiler of this little boat was of a new system invented by the constructor. The jury does not give further details. Walschaerts received the silver medal.

In 1842 Walschaerts was taken into the shops of the State Railway at Malines as a mechanic. Machine tools existed only in the most rudimentary forms, and the store rooms were badly provisioned. The lack of organization in the shops rendered a man of Walschaerts' abilities particularly valuable, and at the end of two years he was made shop foreman at Brussels. Although he was only 24 years of age he had already shown the qualities which make an engineer, which should have carried him in a few years to be the technical head of the motive power department. It is humiliating to be compelled to say that he remained shop foreman throughout his life.

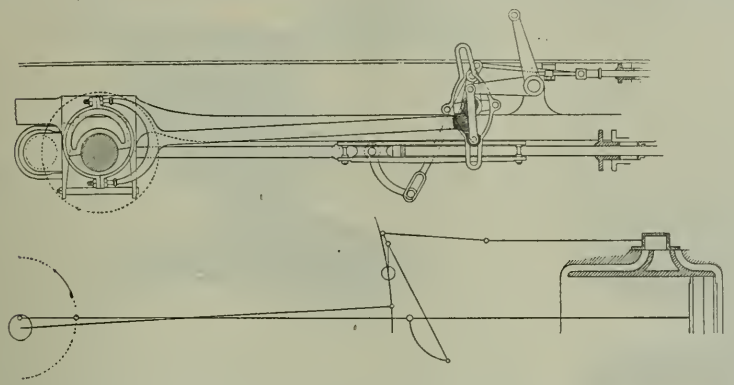


Fig. 1—Walschaerts Valve Motion—Belgian Patent, October 5, 1844.

The first locomotives came from England and had not been in service for more than ten years when Walschaerts was made foreman. The railroad was growing rapidly and it was necessary to increase the forces and to acquire experience. Walschaerts was not content with the duties incurred in these difficult circumstances, but began his career by the invention of his system of valve motion.

On October 5th, 1844, Mr. Fischer, Engineer of the Belgian State Railways filed for Egide Walschaerts an application for a patent relating to a new system of steam distribution applicable to stationary steam engines and to locomotives. This Belgian patent was issued on November 30th, 1844, for a term of fifteen years. The rules of the department did not allow a foreman to exploit a Belgian patent for his own profit and this explains probably the intervention of Mr. Fischer, who has never claimed the slightest part, material or moral, of the invention.

On October 25th of the same year Walschaerts took out a patent in France for the same invention. There also exists among the documents left by the inventor, a contract signed at Brussels in 1845 by Demeuldre, from which it appears that he undertook to obtain a patent of importation into Prussia for the new valve motion, subject to an assignment by Walschaerts of half of the profits to be deducted from the introduction of the new valve motion in this country. It is probable, however, that this contract was never carried out.

The design attached to the Belgian patent is reproduced in Figure 1 herewith. In this primitive arrangement the link oscillated on a fixed shaft, in regard to which it was symmetrical, but it had an enlarged opening at the center so that only at the ends was it operated without play by the link block, which was made in the form of a simple pin. There was only one eccentric, the rod of which terminated in a short T carrying two pins. The reverse shaft operated the eccentric rod and maintained it at the desired height. For one direction the lower pin of the T engaged in the lower end of the link, and to reverse the engine the rod was raised so that the upper pin engaged in the upper end of the link. The angle of oscillation of the link varied with the position of the pin in the link, and this oscillation was transmitted by an arm to the combining lever, which was also operated by the crosshead.

The central part of the link could not be used for the steam distribution, as it was necessary to enlarge it to allow for the play of the pin which was not in operation. It may be asked why the inventor used two separate pins mounted on a crosspiece on the end of the eccentric rod instead of a single pin on the center of the rod which would have served for both forward and backward motion without requiring the center enlargement of the link. It must be borne in mind that the raising or lowering of the eccentric rod by the reverse shaft was equivalent to a slight change in the angular advance of the eccentric. Consequently with the link of a sufficient length to keep down the effect of the angularity it was necessary to reduce as much as possible the movement of the eccentric rod. Notwithstanding its differences the mechanism described in the patent of 1844 is in principle similar to the valve motion with which every one is to-day familiar, and which the inventor constructed as early as 1848, as is shown by a drawing taken from the records of the Brussels shops, on which appears the inscription "Variable expansion; E. Walschaerts' system applied to Locomotive No. 98, Brussels, September 2, 1848."

Figure 2, taken from this drawing, shows the valve motion as we know it to-day. For although it is true that the link and the combining lever are usually placed in a different position so as to shorten the eccentric rod and the valve stem, yet the design of the locomotive often requires an arrangement similar to that shown herewith. The system which Mr. Heusinger von Waldegg invented in 1849, and which he applied in 1850 to 1851 differs only in a few insignificant particulars from that shown in Figure 2. Walschaerts had therefore preceded him.

There are now innumerable systems of steam distribution and the progress made in kinematics has greatly facilitated their development. At the time, however, of Walschaerts' work there was only one system of valve motion which was at all extensively used, namely that of Sharp, with two eccentrics with forked rods. The link attributed to Stephenson was invented by Howe in 1843, and it is doubtful whether Walschaerts had ever seen it. A problem which nowadays appears very simple was for the investigators of that time extremely complicated, and the man who discovered the most correct solution which has yet been put forward merits unreserved admiration.

Notwithstanding his work at the Brussels shops, where he built, with limited equipment, several new locomotives, Walschaerts did not lose sight of the problem of steam navigation, and appeared at the National Exposition at Brussels in 1847 with a screw yacht which ran successfully. The propeller had several blades, each one with only a small part of the thread similar to those of the modern screws. Walschaerts had invented this arrangement himself, without knowing of the results obtained by

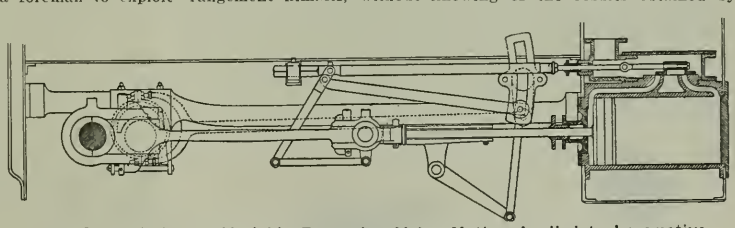


Fig. 2—Walschaerts Variable Expansion Valve Motion Applied to Locomotive No. 98, Brussels, Sept. 2, 1848.

Normand with the Corse in 1841 or 1842. Locomotive practice had taught him the intimate correlation which should exist between the engine and the boiler and in this machine he made use of a boiler of high power which attracted much attention. A similar machine was built a little later by the Couillet Co., as is shown by the plan drawn up in 1853. This machine is compact and light. The documents which remain do not mention the pressure or the speed. It was non-condensing, and in all probability the exhaust steam assisted the draft.

During the years which followed Walschaerts' activity was given entirely to his duties as foreman and it is difficult to determine his part, which was valuable though anonymous, in the design of the railroad equipment. He is credited with the differential throttle in which the opening of an auxiliary slide on the back of the main slide assisted the opening of the valve. He also designed a brake with shoes acting on the rails which was used for a long time in switching locomotives and in which the principle of a lever acting near its dead point was applied in an ingenious manner.

It was very remarkable that his initiative spirit did not suf-

Slide Valve vs. Piston Valve.*

Piston valves in locomotives were used as long ago as 1833, and since that time there have been revivals at various periods; but like some vaccinations, they did not seem to take—until within the last few years. Whether they are to become a fixture and force the D slide valve to the museum remains to be seen. Inquiries of leading roads using large numbers of piston valves, fail to bring out an expression as to which type of valve is doing the better work, the slide or the piston valve. They will say that they are both doing good work when properly designed. There are two types of the piston valve: inside and outside admission. Some are solid and some hollow. It is this modification that brings about the difference of opinion as to their respective merits.

Many broken frames and cylinders have been attributed to the piston valve. Had the trouble been looked into more carefully, they would have found some weak points in the construction, no provision having been made for special relief in case of over-pumping, condensation or careless handling of the engine.

Most modern piston valve engines are of the heavy type with cylinders much larger than were formerly used, and in a great many cases the enginemen have not taken the care of them that they did of the smaller engines; but after the engines have been in service for a while the men become accustomed to them, and the trouble generally disappears. I can see no reason why the frames and cylinders on piston valve engines, if properly designed, are subject to breakage any more than the slide valve engines, if proper provision is made against careless handling. There are four essential points to be considered—cost, maintenance, steam distribution, and fuel economy. There seems to be no question that the first cost of the piston valve is less than the slide valve on simple engines, and where four-cylinder compound types of engines are used, it takes the place of two valves and reduces the motion to that of a simple engine.

With the piston valve we get a better balance of the valve, which makes it easier to handle and decreases the wear and tear on the motion work. With the increased size of engines and steam pressure, the ordinary D balance valve increases in size proportionately, and while we may balance a slide valve in the same ratio as the valves on smaller engines, the difference in the unbalanced surface increases with the size of the engine and this increases the wear on the valve, link motion and eccentric straps, and increases the work necessary on the part of the engineer to handle the engine. This being a fact, I have experienced a great deal of trouble keeping the valves on our slide valve engines square, while on the other hand we do not experience trouble of this kind with the piston valve until after the engine has been out of the shop for a long while and the parts become badly worn. With the use of the inside admission piston valve we do away with the metallic valve stem packing, which means a great saving, as we only have the exhaust pressure on the packing side, and the fibrous packing answers the purpose and lasts a long while. With the slide valve on large engines we can hardly exceed 25,000 miles before the valves need facing, and ofttimes sooner than that. When this has to be done, it means the loss of the use of the engine for a day at least with a cost of $12 to $14 for labor, while with the piston valve if the rings are broken or need attention the valve can be removed, new rings applied in from 30 to 40 minutes, and the engine is ready for service again. No doubt the question will be asked, "Do the bushings ever wear?" Yes, but I have never seen a bushing that will not run from shopping to shopping, and they generally run 200,000 miles without re-boring, which is nearly always done while the engine is in the back shop. Another advantage of the piston valve over the slide valve is the accessibility to its parts. When an engine needs its valves reset after running some time, the port marks on the valve stem become obscured, and possibly the man who is about to do the work has a different tram or wants to get different marks on the stem. With the slide valve engine the machinist has to use the block and tackle and raise the covers of the steam chest before he can make his new marks, while with the piston valve he simply has to remove two plugs on each end of the chest leading directly to the edge of the steam port. This means a saving of time, and time is valuable in a busy shop or roundhouse.

It is claimed by many that what we gain in the maintenance of the piston valve we lose through the leakage of the packing rings. No doubt, there are some leaks through the packing rings in the valve as they or the bushing become worn; but, on the other hand, how many times are slide valves slightly cut and allowed to run in this condition until they wear themselves smooth again? And while they are wearing themselves smooth the engineer will double the amount of oil to the valves and cylinders to prevent the lever from driving him out of the cab.

Another thing in favor of the piston valve is that there are no steam chests to break or gaskets to leak, and all the oil used goes directly to the valve and cylinder. It has been claimed that the piston valve engines ride hard, but often other things are the real cause. Engines set with too much lead will ride hard, or if set too late will cause them to pound. Again, the cylinder and valve may be poorly designed. If it has too little clearance, with a slide valve when compression takes place the valves can rise from the seat and thus relieve compression; but with the piston valve there is no such relief, and this will cause the piston valve engine to ride hard and may do damage unless proper means are provided for relief.

On the other hand, too much cylinder and valve clearance will cause an engine to pound through not having sufficient compression to balance the reciprocating parts, assuming that the cylinder and valve clearance are right. Then a faulty design of piston valve will invariably cause trouble. The inside admission solid valve acts as a piston for each exhaust and takes up the slack in the valve motion, and increases the lead; this is very hard on valve gears and makes the engine ride hard. The difference of pressure on the two ends of this type of valve often amounts to over a ton, for the moment after exhaust takes place the outside admission solid valve becomes unbalanced on the admission side as the steam enters the cylinder, and the high pressure at the opposite end takes up the slack and decreases the lead as the valve gear wears. In the outside admission hollow valve the area of the valve stem unbalances this type to the extent of about 600 lbs. at a 200-lb. boiler pressure, and always in the same direction which causes the engine to go lame as the gear wears. Now with the inside admission hollow valve these defects are absent and the valve is so well balanced that it works easily and requires less power to operate than the slide valve. Tests made by the C., B. & Q. demonstrated that the relative frictional resistance was only about half as much in a piston as in the slide valve.

It is also claimed that the piston valve engine will not run as fast as the slide valve engines. If this is true it is because of faulty design; for an engine intended for fast running the cylinder and valve clearance should be greater than for slow, heavy work. The faster the service the greater should be the cylinder and valve clearance.

Of inside admission piston valves there are two classes, viz.: the solid and built-up types. The solid valves use snap rings, which must of necessity be light section. When steam gets under these rings of light section it causes them to expand into the ports, and unless the corners are well rounded it may cause them to catch and break the valves or rings, or both. On the built-up type the rings may have a heavier cross section and the body of the valve may be cut away so that L-shaped rings may be used, which gives a better admission of steam. Then in case it does break it is only necessary to renew the part that is broken instead of the entire valve. It has also been said that the piston valve engine will not steam as well as the slide valve engine. I can see no reason why this should be so. In regard to the lubrication of the piston valve it has been my experience to find the piston valve much easier to lubricate than the slide valve, and where engines have to drift for a long distance it is found to be a good practice to lower the lever in the quadrant to about half way and to admit just steam enough to the cylinders and valves to carry the oil to the walls of the bushings and cylinders. I think the better practice of oiling the piston valve is to carry the oil to each end of the valve, and as the valve travels back and forth the oil goes directly to the wearing parts. This is done by having a bracket pipe leading to each end of the valve bushing from the main oil pipe. Care must be taken, however, not to drill holes in the bushing so that when the valve is standing in exhaust position at full travel the holes will come in communication with the exhaust port, because then every time the engine exhausts the oil in the pipe would be carried away with the exhaust steam.

With the piston valve we get a much larger port than with the slide valve, and this large opening gives a better admission and release of the steam to and from the cylinder than can be obtained by the slide valve. However, I have been unable to obtain any indicator cards that are comparative; but where the valve gives such a large port opening, both to the steam and exhaust, there seems to be no question but what we get a better admission and release. It has been claimed by some that the steam is wire-drawn on account of the beveled shape of the valve on the admission side; but the increased area of the port overcomes this, as it does not show on indicator cards taken with this style of valve. The modern packing ring is made in an L shape, which gives at least ¼-in. perpendicular opening to the steam port, and helps to give a better admission. With the slide valve the Allen port was introduced to get a better steam admission, and while this did help at a high cut-off, it is practically of no use in starting. While this supplementary port helps the admission of steam to the cylinder, it has no effect upon the release, as these supplementary ports do not come into play while the engine is exhausting. It is also necessary with the Allen ported valve to have the valves set with considerable negative lead in full gear forward motion on account of the lead increasing so fast as the engine is connected that when

*From a paper by L. S. Allen, read at the thirteenth annual convention of the Traveling Engineers' Association, held at Detroit, September 12 to 15.

plugs and sleeves screwed up to a steam tight fit, and all plugs and all caps which make their seat on the end of sleeve should be screwed up tightly, using graphite in the threads, for by the unscrewing of the caps we are allowed to inspect the staybolt proper, when thought advisable.

Tools for installation are shown on tool sheet, also method of removing old rigid water space stays and enlarging the holes in outer sheet for flexible stays.

Referring to tool sheet, Fig. 1 is the method of removing the old bolt by using an enlarged twist drill, following the prick punched center of old bolt, drilling through outer sheet and using a reduced diameter of drill through inner sheet, collapsing the end (C), or separating such by a Wagstaff drill, leaving the hole in outer sheet enlarged ready for roughing reamer for the flexible staybolt sleeve.

Fig. 5 is a butt mill, which enlarges the hole in like manner, from tap drill size.

Fig. 2 is a roughing reamer (B), with guide bar (A), to be used in air drill to quickly straighten and size the hole in outer sheet.

Fig. 3 is a finishing reamer, with guide bar, for exactly sizing each hole previous to tapping.

Fig. 4 is a taper tap with guide bar, ¾ inch taper to the foot, 12-V threads, for finish tapping the hole in outer sheet to receive the sleeve.

Taper tapping differs from straight tapping. The reamed hole should always be of exact size for the tap to give good full threads. The tap should not be used to enlarge the hole, for this process is both unmechanical and detrimental to the tap.

A straight tap of the right diameter used before the taper tap, facilitates the tapping operation and relieves the taper tap from great strain and clogging.

Fig. 6 shows the method of tapping the inner sheet with the regular staybolt tap to aline with the installed sleeve. A cap (B) of the style (D) flexible stays is used as a means for holding the bushing (C), which holds the shank of the tap in line.

The sleeve is screwed into the outer sheet by the stud nut shown in Fig. 8, and as it takes nearly two turns to make a steam-tight fit, after the sleeve just fills the tap hole, it is necessary that the tapping operation should be gaged accordingly.

Fig. 7 shows the style (D) both installed ready for riveting over the end (F). B is a cap such as is used for the flexible stay, cut away for the admission of a dolly bar, bucking tool or holder-on. This cap protects the sleeve face where the cap makes a steam-tight joint during the riveting operation.

The dolly bar or holder-on is made of a piece of axle, 4 or 5 in. in diameter. C is a tool steel plug inserted therein, hardened and tempered.

The style of bolt here shown can be put in place with an alligator wrench on the firebox end or by means of a slot in the head.

Flexible staybolts should not be installed alternately, which has been done in some cases, for it is but a short time before the rigid water space stay has to be taken out and the flexible stay incorporated to effect a sensible installation.

There is no rule to apply regarding the selection of the installation for flexible water space stays other than to cover completely the natural breaking zone of each style fire-box.

The cost of flexible stays range from 40 cents to 70 cents a piece, complete, with bolt.

The cost of installing the flexible water space stay with proper tool equipment and systematic methods should not exceed 25 cents per bolt, and when installed in large quantities the cost should be as low as 15 cents per bolt.

In comparing the relative cost of maintaining a fire-box for four years having rigid staybolts in which the breakage amounted to 40 per cent. with a similar box in which 40 per cent. of the bolts were flexible the author showed that the outlay for staybolts in the former would be $600 as against $328 for the latter.

The stresses which tend to rupture a staybolt are:—That which operates to tear the bolt apart, the tensile stress; that tendency to break it, transversely or cross wise, due to expansion of fire sheet, the transverse stress; and that due to shear which tends to strip the threads from the bolt and hole in the sheet.

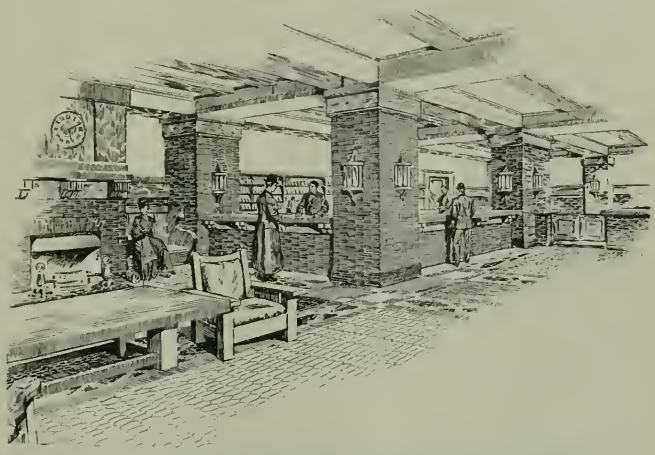
The fact that staybolt breakages have in no sense diminished, regardless of the quality of iron used and the many modifications of the forms devised in the rigid stay in the effort to provide flexibility, notwithstanding that water spaces have been widened, leads us to conclude that the fire-box as now constructed is too rigidly stayed to allow of economic and safe working where cost of maintenance of the complete machine is more or less affected in consequence of the expense accruing from the fire-box and staybolt charges of repairs incident to the constant disintegration and destruction of material, the result of shock, strain, vibration, corrosion and heat.

The force which works on the fire sheet in its course of expansion, due to high temperatures of furnace heat, throws a stress on the uttermost fibers of the rigid staybolt far in excess of the tensile stress, and as it is a reversal or vibratory stress the effect on the structure of the staybolt iron is too severe to warrant safe conclusions as regards maintaining a reasonable factor of safety.

The transverse stress breaks the rigid staybolt, not the tensile stress, and to enable the expansive forces to take their natural course with least resistance the flexible staybolt has been designed as a water space stay, as the most perfect means of affording and maintaining flexibility under all conditions of fire-box service, adding to the life of both sheet and staybolt.

New City Ticket Office of the Alton at Chicago.

A short time ago the Chicago & Alton moved its Chicago city ticket office into new quarters in the Rector Building at the corner of Clark and Monroe streets. The design and arrangement of the office is quite unique in character, some idea of which is conveyed by the accompanying engraving. The idea dominating the interior design was to give to it the appearance of a comfortable waiting room and office in a railroad station, the treatment of the walls, the design of the furniture and the general arrangement being in accordance with this idea. The color scheme harmonizes with the green granite base and the Bedford stone of the exterior of the building. The woodwork is weathered oak of a gray tone and the wainscot and piers are red brick laid in white mortar.



New City Ticket Office of the Alton at Chicago.

The ceiling is sand float plaster finished in white enamel and treated with heavy oak beams. The floor is of vitreous tile in colors to harmonize with the rest of the interior. There is a commodious fire-place with logs, andirons and built in seats, a large table with top 8 in. thick, and numerous easy chairs, all in Mission style. Above the fire-place is a clock with delft dial and white enamel hands and figures. The metal work of the electric light fixtures is finished in verde antique bronze, the fixtures being of special design. Other features of the office are in keeping with those described, the general effect of the interior being most pleasing.

Railroad Decisions in July.

The following decisions in railroad cases were handed down by the United States Supreme Court and the Federal courts during July:

Protection of cars undergoing repairs.—A car repairer assumes the risk of injury by working under a car liable to be moved by contact with other cars, where he fails to protect himself by signals as required by a rule of the company. Canadian Pacific Railway Co. vs. Elliot, 137 Fed. Rep. 904.

Liability of lessor railroad.—The Ohio statue requiring a railroad company leasing its railroad jointly liable with the lessee on all right of action accruing to any one for any negligence or default growing out of the operation or maintenance of the road or in any wise connected therewith applies only to liabilities growing out of its duties as a carrier and not out of its duties as an employer. Beltz vs. Baltimore & Ohio R. Co., 137 Fed. Rep. 1,016; Axline vs. Toledo, Walhonding Valley & Ohio R. R. Co., Fed. 169.

Fellow servants.—A railroad section hand working on a rail-

is unlawful and not otherwise, suspend the operation of the order during pendency of the proceeding or until the further order of the Court. If no proceedings to vacate an order are begun within the time limit or if the effect of the order has not been suspended in any of the ways provided, it becomes known as a final order, and carriers who knowingly neglect to obey a final order shall forfeit to the United States $5,000 for each offence, or $5,000 a day for a continuing violation. The Commission is entitled to apply to the Circuit Court in its own name for an enforcement of an order, and in such a case the Court can enforce obedience by a writ of injunction. It is also provided that copies of schedules and tariffs and the statistics contained in the annual reports of carriers made to the Commission, as required by the Act, shall be preserved as public records in the custody of the Secretary of the Commission and shall be received as *prima facie* evidence of what they purport to be.

A new section is added to the Act to Regulate Commerce following Section 16 and numbered 16A which provides that after a decision has been made by the Commission any party thereto may at any time make application for a re-hearing, to be granted by the Commission at its discretion.

Section 20 of the Act to Regulate Commerce is amended to include the owners of all railroads engaged in interstate commerce as defined in the Act among those from whom annual reports are required. It is also provided that accidents to passengers, employees and other persons shall be reported, together with the causes of them. It is provided that the report shall be for the year ending June 30 and must be filed by September 30 next following, with penalties for non-compliance. The Commission also has authority to require the carriers to file monthly reports of earnings and expenses within a specified period. The Commission may in its discretion prescribe the forms of accounts to be kept by carriers, including traffic movement as well as receipts and expenditures of money. The Commission shall at all times have access to all accounts, records and memoranda kept by carriers subject to the Act, and it shall be unlawful for such carriers to keep any other accounts, records or memoranda than those prescribed or approved by the Commission, which is also entitled to employ special agents or examiners. A severe penalty is laid on any person who shall wilfully make a false entry in accounts, records or memoranda kept by a carrier, or who shall wilfully destroy or by any other means falsify these records. The examiners are estopped by a heavy forfeit from divulging any fact that may come to their knowledge during examination except in so far as they may be directed by the Commission.

Setting Valves with the Walschaert Gear.

BY THOMAS F. CRAWFORD, M.E.

The following method of setting locomotive valves with the Walschaert gear is in use at the Baldwin Locomotive Works:

The first step is to connect all rods and pins with the exception of the eccentric rod. This rod joins the crank pin arm to the base of the link and is left disconnected so that the valve may be moved freely backward and forward in obtaining the various port tram marks on the valve-stem. These are obtained in the

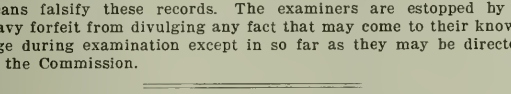


Fig. 1. *Fig. 2.*

Tram Marks on Valve Stem.

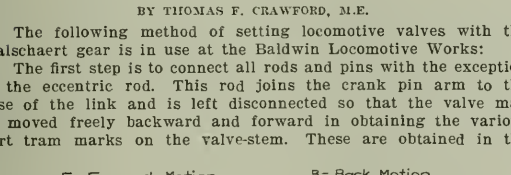


Fig. 3—Walschaert Valve Gear at Half Stroke.

usual manner, as are also the dead center marks on the rim of the main drivers. The engine, in this case a Pennsylvania consolidation, was equipped with inside admission piston valves and was therefore supplied with special holes through the steam chest walls so that the port edges of the valve could be placed exactly square with the admission port edge.

The next step is to connect the eccentric rods on both sides and proceed to catch in succession the various centers for both the forward and back motions with the usual tram marks on the valve stem. For example, suppose the valve tram marks on the stem are as shown in Fig. 1; the back motion lapping the forward on both ends by the amount indicated. This means that the eccentric rod must be shortened, but the amount is the important question. At present it is customary to subtract the differences between the back and front marks at either end and shorten the rod three-quarters of this amount. Difference at front end, ¼ in.; difference at back end, ⅛ in.; ¼ in. — ⅛ in. = ⅛ in.; ¾ of ⅛ in. = $^3/_{32}$ in. The rod would therefore be shortened $^3/_{32}$ in. In another case if the marks were as in Fig. 2 the differences would be added and three-quarters taken. The three-quarters is obtained by allowing one-quarter for lost motion in the pins; this with the one-half required for the alteration adds to three-quarters.

A peculiar feature of the Walschaert motion is noticed at this point when making the above changes. The forward motion is direct and the back motion indirect owing to the fact that the link is stationary. When the link block is at the bottom of the link in the forward motion the eccentric rod acts direct on the valve rod through the block; but when the latter is at the top of the link in the backward motion the link acts as a rocker. After adjusting the forward and back motion in this manner it is still evident that the distance from pop mark to tram mark is not equalized; in other words, the lead is not the same at each end of the stroke. This inequality is adjusted by means of two large nuts which turn on the valve stem and face against the valve stem crosshead. We now have the valves set with the reverse lever in both corners, that is, with a link block clearance of ½ in. at either end.

It is desired, however, to obtain the most perfect adjustment for the position in which the reverse lever will be worked during the service of the engine. For this reason the link block is raised (in this case 6¼ in.), and the extreme valve travel is marked. If this varies at either end the valve stem nuts are again adjusted, sacrificing their former equality in order to be perfect at this cut-off.

It is well to mention the fact that the Walschaert motion has constant lead no matter what the cut-off may be. The shop man is therefore relieved from that difficult feature of ordinary valve setting—obtaining the proper lead. When the engine leaves the drawing board it has its lead and this remains unchanged. It is also interesting to note that the "kicker," or portion of the gear attached to the main crosshead, does not play any part in the valve setting. Its sole duty is to open the steam port rapidly at the beginning of each stroke.

From the foregoing description it is readily seen that there is a large amount of time saved in setting a Walschaert valve motion as compared with the Stephenson. In one observed case, with the ordinary run of luck it required four men 65 minutes to complete the whole operation. This included taking the left eccentric rod to the smith fire to be shortened $^1/_{32}$ in. In this gang two men barred wheels, one man trammed the main drivers, and the other worked at the valve stem.

The above engine had the following dimensions:

Valve travel	6 in.	Cut off	½ stroke
Stroke of pistons	28 in.	Lead	3-16 in.
Diameter of cylinders	22 in.		
Outside lap	⅛-in. neg.		
Inside lap	⅞ in.		

Foreign Rails for Harriman Lines.

The Harriman lines are about to receive a large order of foreign rails. Twenty-one thousand tons, the contract for which was placed during the early part of November with Spanish mills—the Altos Hornos de Viscaya—are intended for use in California. The contract price for the rails, delivered, including import duty, is around $27 a ton. The prevailing price for domestic rails at the mills is $28, and to carry them to the Pacific Coast the total cost would be almost $38. Delivery of the Spanish rails is to be completed by January next. Five thousand tons are now on their way by chartered steamer to Galveston, Texas, and the balance will be shipped to San Diego, Cal.

For the Southern Pacific's extensions in Mexico a rail contract has been placed with the Monterey Iron & Steel Company. The closing of this contract with a Mexican concern is due largely, if not entirely, to the recent tariff imposed. The Mexican price was around $33 a ton, whereas the existing quotations for American rails laid down in Mexico is upwards of $40, the duty being $10 a ton, while the freight charges, etc., are almost $5. Another Mexican road desired to secure a lot of 14,000 tons of rails from the Monterey people, but it was found impossible to supply them within any reasonable time, as the Monterey mills will be busy for some time to come on the Southern Pacific contract. The order has therefore been placed with the Altos Hornos de Bilbao, one of the principal mills in Spain. The price is a shade above $22